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Method for programming memory

Abstract

A method includes setting a current level of a write signal to a first non-zero value for a first period of time. The write signal is provided to a memory element during the first period of time. The current level of the write signal is adjusted from the first non-zero value to a second non-zero value, different from the first non-zero value, for a second period of time. The write signal is provided to the memory element during the second period of time. The current level of the write signal is adjusted from the second non-zero value to a third value, different from the first non-zero value and different from the second non-zero value, for a third period of time. The write signal is provided to the memory element during the third period of time.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 18/077,580, filed on Dec. 8, 2022, which is a Divisional of U.S. application Ser. No. 17/199,849, filed on Mar. 12, 2021 (now U.S. Pat. No. 11,527,289, issued on Dec. 13, 2022). The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND
(1) Many modern day electronic devices contain electronic memory. Electronic memory may be volatile memory or non-volatile memory. Non-volatile memory is able to store data in the absence of power, whereas volatile memory is not. Some examples of next generation electronic memory include magnetoresistive random-access memory (MRAM), resistive random-access memory (RRAM), phase-change random-access memory (PCRAM), and conductive-bridging random-access memory (CBRAM).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1**A illustrates a diagram of some embodiments of a memory array comprising a plurality of memory cells.
(3) FIG. **1**B illustrates a graph of some embodiments of a first signal and a reference signal for programming the plurality of memory cells of FIG. **1**A.
(4) FIG. **2** illustrates a graph of some embodiments of a second signal for programming memory cells.
(5) FIG. **3** illustrates a graph of some embodiments of a third signal for programming memory cells.
(6) FIG. **4** illustrates a graph of some embodiments of a fourth signal for programming memory cells.
(7) FIG. **5** illustrates a graph of some embodiments of a fifth signal for programming memory cells.
(8) FIGS. **6-9** illustrate graphs of some embodiments of a source line voltage, a bit line voltage, a word line voltage, a memory element resistance, and a current passing through the memory element, respectively.
(9) FIGS. **10-12** illustrate flow diagrams of some embodiments of methods for programming memory elements.
DETAILED DESCRIPTION
(10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of

simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(12) Many modern integrated chips include memory arrays. For example, a memory array comprises a plurality of bit lines and a plurality of word lines coupled to a plurality of memory elements arranged in a plurality of rows and columns. Some memory arrays may also comprise one or more source lines coupled to the plurality of memory elements. A first memory cell of the memory array comprises a first memory element coupled to a first transistor configured to selectively provide access to the first memory cell. The first memory element and the first transistor are coupled in series between a first bit line and a first source line. Further, a gate of the first transistor is coupled to a first word line.
(13) During operation of the memory array, the first memory cell may undergo a programming process. For example, the first memory cell may undergo a write operation to in order to store a bit in the memory cell (e.g., to set the state of the memory cell to a logic “1” or a logic “0”). For example, when the first memory element comprises a magnetic tunnel junction (MTJ), a write operation comprises passing a current through the MTJ to create a magnetic field in the MTJ. The magnetic field induced in the MTJ sets a polarity of a free layer of the MTJ. The polarity of the free layer of the MTJ relative to a polarity of a reference layer of the MTJ determines the resistance of the MTJ. The resistance of the MTJ determines the logic state of the MTJ.
(14) The current may have a single magnitude that is maintained for the duration of time in which the current is passed through the memory element. The duration of the current multiplied by the magnitude of the current must be greater than or equal to a programming threshold value to successfully program the memory element. For example, the current magnitude may be large and the duration may be brief, or vice versa. Because a high current and/or a long period of time may be required to perform the write operation, a performance (e.g., a power efficiency and/or a speed) of write operation may be low.
(15) Various embodiments of the present disclosure are related to a method for programming a memory cell that improves a performance of the programming. The method comprises setting a current level of a write signal to a first value for a first period of time. The write signal is provided to a memory element during the first period of time. The current level of the write signal is adjusted from the first value to a second value, different from the first value, for a second period of time. The write signal is provided to the memory element during the second period of time. The current level of the write signal is adjusted from the second value to a third value, different from the first value and different from the second value, for a third period of time. The write signal is provided to the memory element during the third period of time.
(16) By providing a signal having multiple current levels to the memory element to program the memory element, a performance of the programming (e.g., a speed and/or power consumption of the programming) may be improved. For example, the first value and the second value can be tuned in order to achieve a desired balance between programming duration and power consumption. In some embodiments, by increasing the first value and/or the second value, a duration of the programming may be reduced. In some embodiments, by reducing the first value and/or the second value, a power consumption of the programming may be reduced. Thus, by tuning both the first value and the second value, an optimal balance between programming duration and power consumption may be achieved.

(17) Referring to FIGS. **1**A and **1**B simultaneously, FIG. **1**A illustrates a diagram **100***a* of some embodiments of a memory array comprising a plurality of memory cells (e.g., **102***a*-**102***d*), while FIG. **1**B illustrates a graph **100***b* of some embodiments of a first signal **120** and a reference signal **122** for programming the plurality of memory cells (e.g., **102***a*-**102***d*).
(18) As shown in diagram **100***a* of FIG. **1**A, the memory array comprises a plurality of bit lines **104***a*-**104***b* and a plurality of word lines **106***a*-**106***b* arranged in a plurality of rows and columns. For example, the memory array comprises a first bit line **104***a* and a second bit line **104***b* arranged in columns, and a first word line **106***a* and a second word line **106***b* arranged in rows. In some embodiments, the memory array also comprises a first source line **108***a* and a second source line **108***b*. In some embodiments, the first source line **108***a* is coupled to the second source line **108***b*.
(19) The memory array further comprises a plurality of memory cells **102***a*-**102***d* coupled to the plurality of bit lines **104***a*-**104***b*, the plurality of word lines **106***a*-**106***b*, and the one or more source lines **108***a*-**108***b*. For example, the memory array comprises a first memory cell **102***a* coupled to the first bit line **104***a*, the first word line **106***a*, and the first source line **108***a*. The first memory cell **102***a* comprises a first memory element **110***a* coupled to a first transistor **112***a*. The first memory element **110***a* and the first transistor **112***a* are coupled in series between the first bit line **104***a* and the first source line **108***a*. A gate of the first transistor **112***a* is coupled to the first word line **106***a*.
(20) Similarly, a second memory cell **102***b*, a third memory cell **102***c*, and a fourth memory cell **102***d* comprise a second memory element **110***b*, a third memory element **110***c*, and a fourth memory element **110***d*, respectively, and further comprise a second transistor **112***b*, a third transistor **112***c*, and a fourth transistor **112***d*, respectively.
(21) In some embodiments, the memory array is or comprises a magnetoresistive random access memory (MRAM) array. In such embodiments, the first memory element **110***a* is or comprises a first magnetic tunnel junction (MTJ), the second memory element **110***b* is or comprises a second MTJ, the third memory element **110***c* is or comprises a third MTJ, and the fourth memory element **110***d* is or comprises a fourth MTJ.
(22) In some embodiments, circuitry **114** is coupled to the first bit line **104***a*, the second bit line **104***b*, the first source line **108***a*, the second source line **108***b*, the first word line **106***a*, the second word line **106***b*, or any combination of the foregoing. The circuitry **114** is configured to program the plurality of memory cells **102***a*-**102***d* of the memory array. In some embodiments, the programming comprises writing to one of the memory elements (e.g., **110***a*-**110***d*) to set a state of that memory element. In some embodiments, the circuitry **114** may comprise a bit line driver **114***a* coupled to the plurality of bit lines **104***a*-**104***b* and a word line driver **114***b* coupled to the plurality of word lines **106***a*-**106***b*. In some embodiments, the circuitry **114** may additionally comprise a source line driver **114***c* coupled to the plurality of source lines **108***a*-**108***b*. In some other embodiments, the plurality of source lines **108***a*-**108***b* may be coupled to ground. In some embodiments, the circuitry **114** may comprise a control unit **114***d* configured to provide control signals to the bit line driver **114***a*, the word line driver **114***b*, and/or the source line driver **114***c*. In some embodiments, the control signals control addressing of the bit line driver **114***a*, the word line driver **114***b*, and/or the source line driver **114***c*, to enable the circuitry **114** to access specific memory cells within the memory array.
(23) In some embodiments, the circuitry **114** performs a write operation by passing current through one of the memory elements. For example, in some embodiments, the write operation is performed by operating the word line driver **114***b* to assert a transistor while operating the bit line driver **114***a* to generate a current on a bit line associated with the transistor. For example, a write operation may be performed on the first memory cell **102***a* by operating the word line driver **114***b* to generate a voltage on the first word line **106***a* to turn on the first transistor **112***a*, and by operating the bit line driver **114***a* to generate a first signal on the first bit line **104***a*. As a result, current passes through the first memory element **110***a*, from the first bit line **104***a* to the first source line **108***a* (e.g., from point **116** to point **118**), or vice versa (e.g., from point **118** to point **116**), depending on desired state

of first memory element **110***a*. In other words, in some embodiments, passing current through the first memory element **110***a* in a first direction (e.g., from the first bit line **104***a* to the first source line **108***a*) may set the first memory element to a first state (e.g., a logic “0” state), while passing current through the first memory element **110***a* in a second direction (e.g., from the first source line **108***a* to the first bit line **104***a*) may set the first memory element to a second state (e.g., a logic “1” state).

(24) In some embodiments, the magnitude of the current and the duration of the current used to write the first memory element **110***a* may need to meet some general requirements to successfully set the state of the first memory element **110***a*. For example, in some embodiments, a product of the magnitude of the current passed through the first memory element **110***a* multiplied by the amount of time the current is passed through the first memory element **110***a* may need to be greater than, or equal to, a programming threshold value in order to successfully set the state of the first memory element **110***a*.

(25) As shown in graph **100***b* of FIG. 1B, in some embodiments, the circuitry **114** (e.g., the bit line driver **114***a* and/or the source line driver **114***c*) provides a first signal **120** to the first memory element **110***a* to write the first memory element **110***a*. At a reference start time **124**, the circuitry **114** adjusts a current of the first signal **120** to a first value **138** (e.g., some first non-zero value). The current of the first signal **120** is maintained at the first value **138** until a first time **126** is reached, at which point the circuitry **114** adjusts the current of the first signal **120** to a second value **140** (e.g., some second non-zero value) different from the first value **138** (e.g., greater in magnitude than the first value). The current of the first signal **120** is maintained at the second value **140** until a second time **128** is reached, at which point the circuitry **114** may adjust the current of the first signal **120** to zero.

(26) In other words, the circuitry **114** sets a current level of the first signal **120** to the first value **138** for a first period of time **132**, the circuitry **114** adjusts the current level of the first signal **120** from the first value **138** to the second value **140** for a second period of time **134**, and the circuitry **114** adjusts the current level of the first signal **120** from the second value **140** to a third value (e.g., zero) for at least a third period of time (not labeled). In some embodiments, prior to the first period of time **132** there is no current (e.g., 0 amps) passing through the first memory element **110***a*. In some embodiments, the first value **138** and the second value **140** have a same sign (e.g., are both positive or are both negative). In some embodiments, the first period of time **132** is longer than the second period of time **134**. In some other embodiments, the first period of time **132** is shorter than the second period of time **134**.

(27) The product of the first value **138** multiplied by the first period of time **132**, plus the product of the second value **140** multiplied by the second period of time **134** is greater than, or equal to, the programming threshold value for writing so that the first signal **120** is capable of successfully writing the first memory element **110***a*.

(28) By providing the first signal **120** to the first memory element **110***a* to write the first memory element **110***a*, the time required to write the first memory element **110***a* may be reduced. For example, the time required to write the first memory element **110***a* using the first signal **120** may be less than the time required to write the first memory element **110***a* using a signal that has a single current level (e.g., a reference signal **122**) for a reference period of time **136** which starts at the reference start time **124** and ends at a reference end time **130**.

(29) In some embodiments, because the second value **140** is greater than the first value **138**, and because the current level of the first signal **120** is set to the second value **140** for the second period of time **134**, the second time **128** (e.g., when the second period of time **134** ends) may be less than the reference end time **130** (e.g., when the reference period of time **136** ends) while the first signal **120** still satisfies the programming threshold value. In other words, in some embodiments, the sum of the first period of time **132** plus the second period of time **134** (e.g., the duration of the first signal **120**) is less than the reference period of time **136** (e.g., the duration of the reference signal

122). Thus, the first signal **120** may reduce a time required to program the first memory element **110***a*.
(30) Further, by providing the first signal **120** having multiple current levels to the first memory element **110***a* to program the first memory element **110***a*, a performance of the first signal may be tuned. For example, the first value **138** and the second value **140** can be tuned in order to achieve a desired balance between the time required to program the first memory element **110***a* and the power required to program the first memory element **110***a*. Thus, a performance of the programming may be optimized and/or improved.
(31) In some embodiments, the circuitry **114** may set and/or adjust the current being provided to the first memory element **110***a* by controlling voltages applied to a source (not labeled), drain (not labeled), and gate (not labeled) of the first transistor **112***a*. For example, by applying a first voltage to the source of the first transistor **112***a*, a second voltage to the drain of the first transistor **112***a*, and a third voltage to the gate of the first transistor **112***a* for the first period of time **132**, the first transistor **112***a* may be turned on to a first current level (e.g., the first value **138**). Then, by adjusting the voltage applied to the gate to a fourth voltage for the second period of time **134**, the first transistor **112***a* may be turned on to a second current level different from the first current level (e.g., the second value **140**). Next, by adjusting the voltage applied to the gate to a fifth voltage for the third period of time (not labeled), the first transistor **112***a* may be turned off such that no current flows through the first memory element **110***a*.
(32) Although FIG. **1**A illustrates memory array comprising four memory cells, it will be appreciated that an array comprising some other number of memory cells is also feasible. Further, although the first source line **108***a* and the second source in **108** are illustrated as being coupled together, it will be appreciated that in some other embodiments, the first source line **108***a* and the second source line **108***b* may alternatively be electrically isolated from each other.
(33) Although zero amps is used as a baseline current in the embodiments illustrated in FIG. **1**B, it will be appreciated that in some embodiments, some other current value could be used as a baseline current.
(34) FIG. **2** illustrates a graph **200** of some embodiments of a second signal **202** for programming memory cells.
(35) In some embodiments, circuitry (e.g., **114** of FIG. **1**A) provides the second signal **202** to a memory element (e.g., **110***a* of FIG. **1**A) to write the memory element. At a reference start time **124**, the circuitry adjusts a current of the second signal **202** to a second value **212**. The current of the second signal **202** is maintained at the second value **212** until a first time **204** is reached, at which point the circuitry adjusts the current of the second signal **202** to a first value **138** different from the second value **212** (e.g., lesser in magnitude than the second value **212**). The current of the second signal **202** is maintained at the first value **138** until a second time **206** is reached, at which point the circuitry may adjust the current of the second signal **202** to zero.
(36) In other words, the circuitry sets a current level of the second signal **202** to the second value **212** for a first period of time **208**, the circuitry adjusts the current level of the second signal **202** from the second value **212** to the first value **138** for a second period of time **210**, and the circuitry adjusts the current level of the second signal **202** from the first value **138** to a third value (e.g., zero) for at least a third period of time (not labeled). In some embodiments, prior to the first period of time **208** there is no current passing through the memory element. In some embodiments, the first value **138** and the second value **212** have the same sign.
(37) The product of the second value **212** multiplied by the first period of time **208**, plus the product of the first value **138** multiplied by the second period of time **210** is greater than, or equal to, the programming threshold value for writing so that the second signal **202** is capable of successfully writing the memory element.
(38) FIG. **3** illustrates a graph **300** of some embodiments of a third signal **302** for programming memory cells.

(39) In some embodiments, circuitry (e.g., **114** of FIG. **1**A) provides the third signal **302** to a memory element (e.g., **110***a* of FIG. **1**A) to write the memory element. At a reference start time **124**, the circuitry adjusts a current of the third signal **302** to a second value **312**. The current of the third signal **302** is maintained at the second value **312** until a first time **304** is reached, at which point the circuitry adjusts the current of the third signal **302** to a third value **314** different from the second value **312**. The current of the third signal **302** is maintained at the third value **314** until a second time **306** is reached, at which point the circuitry may adjust the current of the third signal **302** to zero.
(40) In other words, the circuitry sets a current level of the third signal **302** to the second value **312** for a first period of time **308**, the circuitry adjusts the current level of the third signal **302** from the second value **312** to the third value **314** for a second period of time **310**, and the circuitry adjusts the current level of the third signal **302** from the third value **314** to a fourth value (e.g., zero) for at least a third period of time (not labeled). In some embodiments, prior to the first period of time **308** there is no current passing through the memory element. In some embodiments, the second value **312** and the third value **314** have a same sign.
(41) The product of the second value **312** multiplied by the first period of time **308**, plus the product of the third value **314** multiplied by the second period of time **310** is greater than, or equal to, the programming threshold value for writing so that the third signal **302** is capable of successfully writing the memory element.
(42) In some other embodiments, the circuitry may alternatively set the current level of the third signal **302** to the second value **312** first and then subsequently adjust the current level to the third value **314**.
(43) FIG. **4** illustrates a graph **400** of some embodiments of a fourth signal **402** for programming memory cells.
(44) In some embodiments, circuitry (e.g., **114** of FIG. **1**A) provides the fourth signal **402** to a memory element (e.g., **110***a* of FIG. **1**A) to write the memory element. At a reference start time **124**, the circuitry adjusts a current of the fourth signal **402** to a second value **416**. The current of the fourth signal **402** is maintained at the second value **416** until a first time **404** is reached, at which point the circuitry adjusts the current of the fourth signal **402** to a third value **418** different from the second value **416**. The current of the fourth signal **402** is maintained at the third value **418** until a second time **406** is reached, at which point the circuitry adjusts the current of the fourth signal **402** to a fourth value **420** different from the third value **418**. The current of the fourth signal **402** is maintained at the fourth value **420** until a third time **408** is reached, at which point the circuitry may adjust the current of the fourth signal **402** to zero.
(45) In other words, the circuitry sets a current level of the fourth signal **402** to the second value **416** for a first period of time **410**, the circuitry adjusts the current level of the fourth signal **402** from the second value **416** to the third value **418** for a second period of time **412**, the circuitry adjusts the current level of the fourth signal **402** from the third value **418** to the fourth value **420** for a third period of time **414**, and the circuitry adjusts the current level of the fourth signal **402** from the fourth value **420** to a fifth value (e.g., zero) for at least a fifth period of time (not labeled). In some embodiments, the second value **416**, the third value **418**, and the fourth value **420** have a same sign. In some embodiments, each of the periods of time may have different durations.
(46) The product of the second value **416** multiplied by the first period of time **410**, plus the product of the third value **418** multiplied by the second period of time **412**, plus the product of the fourth value **420** multiplied by the third period of time **414** is greater than, or equal to, the programming threshold value for writing so that the fourth signal **402** is capable of successfully writing the memory element.
(47) In some other embodiments, the circuitry may alternatively set the current level of the fourth signal **402** to the fourth value **420** first, may subsequently set the current level of the third value **418**, and may subsequently set the current level to the second value **416**.

(48) Although the fourth signal **402** is illustrated as having three separate current levels (e.g., **416**, **418**, **420**), it will be appreciated that in some other embodiments, the fourth signal **402** may alternatively have some other number of separate current levels.
(49) In some embodiments, by increasing the number of separate current levels that a write signal comprises, a balance between a duration of the write signal and a power consumption of the write signal may be further tuned and/or optimized.
(50) FIG. **5** illustrates a graph **500** of some embodiments of a fifth signal **502** for programming memory cells.
(51) In some embodiments, the fifth signal **502** is similar to the first signal **120** of FIG. **1**B. For example, a current level of the fifth signal **502** is set (e.g., by circuitry **114** of FIG. **1**A) to a first value **138** for a first period of time **514**, the current level of the fifth signal **502** is adjusted from the first value **138** to a second value **524** for a second period of time **516**, and the current level of the fifth signal **502** is adjusted from the second value **524** to a third value (e.g., zero) for at least a third period of time (not labeled). However, in the fifth signal **502** there exists a first transition time **518** (e.g., a rise time from a reference start time **124** to a first time **504**) during which the current level of the fifth signal **502** transitions from zero to the first value **138**, a second transition time **520** (e.g., a rise time from a second time **506** to a third time **508**) during which the current level of the fifth signal **502** transitions from the first value **138** to the second value **524**, and a third transition time **522** (e.g., a fall time from a fourth time **510** to a fifth time **512**) during which the current level of the fifth signal transitions from the second value **524** back to zero.
(52) In some embodiments, the first transition time **518** spans from before the first period of time **514** to a beginning of the first period of time **514**, the second transition time **520** spans from an end of the first period of time **514** to a beginning of the second period of time **516**, and the third transition time spans from an end of the second period of time **516** to a beginning of the third period of time (not labeled).
(53) In some embodiments, the fifth signal **502** is differentiable. In some embodiments, the first signal **120** of FIG. **1**B may be referred to as a theoretical write signal while the fifth signal **502** may be referred to as an experimental write signal.
(54) Although transition periods (e.g., **518**, **520**, **522**) are only illustrated in FIG. **5** with regard to the fifth signal **502**, it will be appreciated that in some embodiments, any of the first signal **120**, the second signal **202**, the third signal **302**, the fourth signal **402**, or some other suitable programming signal may similarly comprise transition periods.
(55) In some embodiments, the circuitry **114** may, for example, comprise a driver circuit, some other hardware, some combination of hardware and software, or some other suitable circuitry. In some embodiments, the transistors (e.g., **112***a*-**112***d*) may, for example, be or comprise metal-oxide-semiconductor field effect transistors or the like. In some embodiments, the current values (e.g., **138**, **140**, **212**, **312**, **314**, **416**, **418**, **420**, **524**) may, for example, be on the order of milliamps. In some embodiments, the times (e.g., **124**, **126**, **128**, **130**, **204**, **206**, **304**, **306**, **404**, **406**, **408**, **504**, **506**, **508**, **510**, **512**) may, for example, be on the order of nanoseconds.
(56) FIG. **6** illustrates graphs **600***a*-**600***e* of some embodiments of a source line (e.g., **108***a* of FIG. **1**A) voltage, a bit line (e.g., **104***a* of FIG. **1**A) voltage, a word line (e.g., **106**A of FIG. **1**A) voltage, a memory element (e.g., **110**A of FIG. **1**A) resistance, and a current passing through the memory element, respectively.
(57) In some embodiments, when programming the memory element, a first current is passed through the memory element until the resistance of the memory element reaches a threshold, at which point a second current different from the first current (e.g., having a lower magnitude than the first current) is then passed through the memory element at least until the memory element reaches a desired resistance. In some embodiments, by providing a current of higher magnitude to the memory element to begin changing the resistance of the memory element and by providing a current of lower magnitude to finish changing the resistance, a power consumption and a speed of

the programming may be improved.
(58) For example, at a first time **602**, the source line voltage is adjusted from a first source line voltage value **608** to a second source line voltage value **610** (e.g., by a source line driver **114***c* of FIG. **1**), the bit line voltage is set to a first bit line voltage value **612** (e.g., by a bit line driver **114***a* of FIG. **1**A), and the word line voltage is adjusted from a first word line voltage value **614** to a second word line voltage value **616** (e.g., by a word line driver **114***b* of FIG. **1**A). As a result, a first current **626** passes from the source line to the bit line through the memory element, which causes the resistance of the memory element to begin to change from a first resistance value **620** toward a second resistance value **624**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the first threshold **622** at a second time **604**.
(59) At the second time **604**, the word line voltage is adjusted from the second word line voltage value **616** to a third word line voltage value **618** that has a lower magnitude than the second word line voltage value **616**. As a result, a second current **628** (e.g., a current of a lesser magnitude) passes from the source line to the bit line through the memory element which causes the resistance of the memory element to continue changing toward the second resistance value **624**. In some embodiments, using the second current **628** causes the resistance to change at a slower rate than when the first current **626** was used. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the second resistance value **624** at a third time **606**.
(60) In some embodiments, the first threshold **622** is a percentage of a difference between the first resistance value **620** and the second resistance value **624**. For example, the first threshold may, for example, be 50% of the difference (e.g., half way between the two), 60% of the difference, 70% of the difference, or some other suitable value. In some embodiments, the first resistance value **620** represents a logic “0” and the second resistance value **624** represents a logic “1”. Although FIG. **6** illustrates changing the state of the memory device from a logic “0” to a logic “1”, it will be appreciated that this threshold based programming operation may also be used when changing the state from a logic “1” to “0”. In some embodiments, the first resistance value **620** and the second resistance value **624** may, for example, be predetermined resistances.
(61) FIG. **7** illustrates graphs **700***a*-**700***e* of some alternative embodiments of a source line (e.g., **108***a* of FIG. **1**A) voltage, a bit line (e.g., **104***a* of FIG. **1**A) voltage, a word line (e.g., **106**A of FIG. **1**A) voltage, a memory element (e.g., **110**A of FIG. **1**A) resistance, and a current passing through the memory element, respectively.
(62) In some embodiments, a magnitude step-down current signal may be used when changing the memory element from a low resistance state to a first resistance state (e.g., when programming a logic “1”) and a magnitude step-up current signal may be used when changing the memory element from a first resistance state to a second resistance state (e.g., when programing a logic “0”), or vice versa. By providing a step-down current to write a first resistance state and a step-up current to write a second resistance state, a performance of the programming (e.g., a speed or power consumption) may be improved.
(63) For example, at a first time **702**, the source line voltage is adjusted from a first source line voltage value **714** to a second source line voltage value **716**, the bit line voltage is set to a first bit line voltage value **718**, and the word line voltage is adjusted from a first word line voltage value **722** to a second word line voltage value **724**. As a result, a first current **732** passes from the source line to the bit line through the memory element, which causes the resistance of the memory element to begin to change from a first resistance value **728** toward a second resistance value **730**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values until a second time **704**.
(64) At the second time **704**, the word line voltage is adjusted from the second word line voltage value **724** to a third word line voltage value **726** that has a lower magnitude than the second word

line voltage value **724**. As a result, a second current **734** (e.g., a current having a lesser magnitude than the first current **732**) passes from the source line to the bit line through the memory element and the resistance of the memory element to continues approaching the second resistance value **730**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the second resistance value **730** at a third time **706**.
(65) At the third time **706**, the word line voltage and the source line voltage are set to the first word line voltage value **722** and the first source line voltage value **714**, respectively. As a result, a third current **740** is passed through the memory element.
(66) At a fourth time **708**, the source line voltage is set to the first source line voltage value **714**, the bit line voltage is adjusted from the first the first bit line voltage value **718** to a second bit line voltage value **720**, and the word line voltage is adjusted from the first word line voltage value **722** to the third word line voltage value **726**. As a result, a fourth current **736** passes from the bit line to the source line through the memory element, which causes the resistance of the memory element to begin to change from the second resistance value **730** toward the first resistance value **728**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values until a fifth time **710**.
(67) At the fifth time **710**, the word line voltage is adjusted from the third word line voltage value **726** to the second word line voltage value **724**. As a result, a fifth current **738** (e.g., a current having a lesser magnitude than the fourth current **736**) passes from the bit line to the source line through the memory element and the resistance of the memory element to continues approaching the first resistance value **728**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the first resistance value **728** at a sixth time **712**.
(68) At the sixth time **712**, the word line voltage and the bit line voltage are set to the first word line voltage value **722** and the first bit line voltage value **718**, respectively. As a result, the third current **740** is passed through the memory element.
(69) FIG. **8** illustrates graphs **800***a*-**800***e* of some alternative embodiments of a source line (e.g., **108***a* of FIG. **1**A) voltage, a bit line (e.g., **104***a* of FIG. **1**A) voltage, a word line (e.g., **106**A of FIG. **1**A) voltage, a memory element (e.g., **110**A of FIG. **1**A) resistance, and a current passing through the memory element, respectively.
(70) In some embodiments, when the memory array is operating in a standard power mode, currents having higher magnitudes and shorter durations are used to program the memory element, and when the memory array is operating in a low power mode, currents having lower magnitudes and longer durations are used to program the memory element. Thus, quick programming speed may be given priority over low power consumption during standard power mode while low power consumption may be given priority over quick programming speed during low power mode. As a result, a performance of the memory array may be improved. For example, if operating on battery power, a battery life of the battery powering the memory array may be improved when operating in low power mode (e.g., because power is proportional to current squared and thus lower currents correspond to lower power).
(71) For example, a standard power mode is employed during a first period of time **852** and a low power mode is employed during a second period of time **854**.
(72) At a first time **802**, the source line voltage is adjusted from a first source line voltage value **816** to a second source line voltage value **818**, the bit line voltage is set to a first bit line voltage value **820**, and the word line voltage is adjusted from a first word line voltage value **824** to a second word line voltage value **826**. As a result, a first current **840** passes from the source line to the bit line through the memory element, which causes the resistance of the memory element to begin to change from a first resistance value **836** toward a second resistance value **838**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values

until a second time **804**.
(73) At the second time **804**, the word line voltage is adjusted from the second word line voltage value **826** to a third word line voltage value **828** that has a lower magnitude than the second word line voltage value **826**. As a result, a second current **842** passes from the source line to the bit line through the memory element and the resistance of the memory element to continues approaching the second resistance value **838**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the second resistance value **838** at a third time **806**.
(74) At the third time **806**, the word line voltage and the source line voltage are set to the first word line voltage value **824** and the first source line voltage value **816**, respectively. As a result, a third current **850** is passed through the memory element.
(75) At a fourth time **808**, the source line voltage is set to the first source line voltage value **816**, the bit line voltage is adjusted from the first bit line voltage value **820** to a second bit line voltage value **822**, and the word line voltage is adjusted from the first word line voltage value **824** to a fourth word line voltage value **830**. As a result, a fourth current **844** passes from the bit line to the source line through the memory element, which causes the resistance of the memory element to begin to change from the second resistance value **838** toward the first resistance value **836**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values until a fifth time **810**.
(76) At the fifth time **810**, the word line voltage is adjusted from the fourth word line voltage value **830** to a fifth word line voltage value **832**. As a result, a fifth current **846** passes from the bit line to the source line through the memory element and the resistance of the memory element to continues approaching the first resistance value **836**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values until a sixth time **812**.
(77) At the sixth time **812**, the word line voltage is adjusted from the fifth word line voltage value **832** to a sixth word line voltage value **834**. As a result, a sixth current **848** passes from the bit line to the source line through the memory element and the resistance of the memory element to continues approaching the first resistance value **836**. The source line voltage, bit line voltage, and word line voltage are maintained at the aforementioned values at least until the resistance of the memory element reaches the first resistance value **836** at a seventh time **814**.
(78) At the seventh time **814**, the word line voltage and the bit line voltage are set to the first word line voltage value **824** and the first bit line voltage value **820**, respectively. As a result, the third current **850** is passed through the memory element.
(79) In some embodiments, the fourth current **844**, the fifth current **846**, and the sixth current **848** have lower magnitudes than the first current **840** and the second current **842**. In some embodiments, a time between the fourth time **808** and the seventh time **814** is longer than a time between the first time **802** and the third time **806**. Thus, the programming during the first period of time **852** (e.g., when in standard power mode) is quicker and may use more power than the programming during the second period of time **854** (e.g., when in low power mode).
(80) FIG. **9** illustrates a cross-sectional view **900** of some embodiments of a memory cell (e.g., **102***a* of FIG. **1**A).
(81) In some embodiments, a first source/drain region **904***a* and a second source/drain region **904***b* are arranged along a substrate **902**. A gate **906** extends along the substrate **902** between the first source/drain region **904***a* and the second source/drain region **904***b*. A gate dielectric **908** separates the gate **906** from the substrate **902**.
(82) A conductive source line **912** is arranged over the second source/drain region **904***b*. A first contact **910** extends from the conductive source line **912** to the second source/drain region **904***b*. A conductive word line **916** is arranged over the gate **906**. A second contact **914** extends from the conductive word line **916** to the gate **906**.
(83) A landing pad **920** is arranged over the first source/drain region **904***a*. A third contact **918**

extends between the landing pad **920** and the first source/drain region **904***a*. A memory element **924** (e.g., an MTJ or the like) is arranged over the landing pad **920**. In some embodiments, the memory element **924** comprises a magnetic reference layer **926**, a magnetic free layer **930**, and a tunnel barrier layer **928** between the magnetic reference layer **926** and the magnetic free layer **930**. A bottom electrode **922** extends from the memory element **924** to the landing pad **920**. The conductive bit line **934** is arranged over the memory element **924**. A top electrode **932** extends from the conductive bit line **934** to the memory element **924**.

(84) FIG. **10** illustrates a flow diagram of some embodiments of a method **1000** for programming a memory element.

(85) At **1002**, a current level of a write signal is set to a first value for a first period of time. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1002**.

(86) At **1004**, the write signal is provided to a memory element during the first period of time. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1004**.

(87) At **1006**, the current level of the write signal is adjusted from the first value to a second value, different from the first value, for a second period of time. In some embodiments, the current level of the write signal is adjusted to the second value when a resistance of the memory element reaches a threshold. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1006**.

(88) At **1008**, the write signal is provided to the memory element during the second period of time. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1008**.

(89) At **1010**, the current level of the write signal is adjusted from the second value to a third value, different from the first value and different from the second value, for a third period of time. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1010**.

(90) At **1012**, the write signal is provided to the memory element during the third period of time. FIG. **1**B and FIG. **6** illustrate graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1012**.

(91) FIG. **11** illustrates a flow diagram of some embodiments of a method **1100** for programming a memory element.

(92) At **1102**, apply a first voltage at a source of the transistor, a second voltage at the drain of the transistor, and a third voltage at the gate of the transistor for a first period of time. FIG. **1**A and FIG. **6** illustrate a diagram **100***a* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1102**.

(93) At **1104**, apply the first voltage at a source of the transistor, the second voltage at the drain of the transistor, and a fourth voltage, different from the third voltage, at the gate of the transistor for a second period of time. FIG. **1**A and FIG. **6** illustrate a diagram **100***a* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1104**.

(94) At **1106**, apply a first voltage at the source of the transistor, the second voltage at the drain of the transistor, and a fifth voltage, different from the third voltage and the fourth voltage, at the gate of the transistor for a third period of time. FIG. **1**A and FIG. **6** illustrate a diagram **100***a* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1106**.

(95) FIG. **12** illustrates a flow diagram of some embodiments of a method **1200** for programming a memory element. While methods **1000**-**1200** are illustrated and described as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments

of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(96) At **1202**, a first current is passed through a memory element for a first period of time, the first current having a first magnitude and a first sign. FIG. **1**B and FIG. **6** illustrate a graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1202**.

(97) At **1204**, a second current is passed through the memory element for a second period of time, the second current having the first sign and having a second magnitude different from the first magnitude. In some embodiments, the second current is passed in response to a resistance of the memory element reaching a threshold. FIG. **1**B and FIG. **6** illustrate a graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1204**.

(98) At **1206**, a third current is passed through the memory element for a third period of time, the third current having a third magnitude different from the first magnitude and the second magnitude. FIG. **1**B and FIG. **6** illustrate a graph **100***b* and graphs **600***a*-**600***e* of some embodiments corresponding to act **1206**. In some embodiments, the third magnitude has the first sign. In some other embodiments, the third magnitude has no sign (e.g., is zero).

(99) Thus, the present disclosure relates to a method for programming a memory cell that improves a performance of the programming.

(100) Accordingly, in some embodiments, the present disclosure relates to a method. The method comprises setting a current level of a write signal to a first non-zero value for a first period of time. The write signal is provided to a memory element during the first period of time. The current level of the write signal is adjusted from the first non-zero value to a second non-zero value, different from the first non-zero value, for a second period of time. The write signal is provided to the memory element during the second period of time. The current level of the write signal is adjusted from the second non-zero value to a third value, different from the first non-zero value and different from the second non-zero value, for a third period of time. The write signal is provided to the memory element during the third period of time.

(101) In other embodiments, the present disclosure relates to a method for programming memory. The method comprises passing a first current through a memory element for a first period of time, the first current having a first magnitude and a first sign. The method comprises passing a second current through the memory element for a second period of time, the second current having the first sign and having a second magnitude different from the first magnitude. The method further comprises passing a third current through the memory element for a third period of time, the third current having a third magnitude different from the first magnitude and the second magnitude.

(102) In yet other embodiments, the present disclosure relates to a circuit. The circuit comprises a memory array. The memory array comprises a first memory cell coupled to a first bit line, a first source line, and a first word line. The first memory cell comprises a first memory element. Circuitry is coupled to the first memory element and is configured to change a logic state of the first memory element by providing a write signal to the first memory element. The write signal has a first current for a first period of time, a second current for a second period of time, and a third current for a third period of time.

(103) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A circuit comprising: a memory array comprising a memory cell coupled to a bit line, a source line, and a word line, wherein the memory cell comprises a memory element coupled between the bit line and the source line; and circuitry coupled to the memory element and configured to change a logic state of the memory element by providing a write signal to the memory element and setting the write signal to a first current level having a first magnitude and a first sign for a first period of time, a second current level having a second magnitude, different than the first magnitude, and the first sign for a second period of time, and a third current level having a third magnitude, different than the first magnitude and the second magnitude, for a third period of time.
2. The circuit of claim 1, wherein the second period of time is different than the first period of time, and the third period of time is different than the second period of time and the first period of time.
3. The circuit of claim 1, wherein the second magnitude is greater than the first magnitude.
4. The circuit of claim 1, wherein the third current level has the first sign, and the third magnitude is greater than the second magnitude.
5. The circuit of claim 1, wherein the second magnitude is less than the first magnitude.
6. The circuit of claim 1, wherein the first magnitude, the second magnitude, and the third magnitude are non-zero, and the third current level has the first sign.
7. The circuit of claim 1, wherein the circuitry is configured to set the write signal to the second current level in response to a resistance of the memory element reaching a threshold.
8. The circuit of claim 1, the circuitry comprising a bit line driver coupled to the bit line, a source line driver coupled to the source line, and a word line driver coupled to the word line.
9. The circuit of claim 8, the bit line driver configured to control a voltage on the bit line, the source line driver configured to control a voltage on the source line, and the word line driver configured to control a voltage on the word line.
10. The circuit of claim 8, the circuitry further comprising control circuitry coupled to, and configured to control, the bit line driver, the source line driver, and the word line driver.
11. A circuit comprising: a memory array comprising a memory cell coupled to a bit line, a source line, and a word line, wherein the memory cell comprises a memory element coupled between the bit line and the source line; and circuitry coupled to the memory element and configured to change a resistance of the memory element from a first resistance to a second resistance, different than the first resistance, by setting a current through the memory element to have a first magnitude and a first sign for a first period of time, a second magnitude, different than the first magnitude, and the first sign for a second period of time in response to the resistance of the memory element reaching a threshold resistance, and a third magnitude, different than the first magnitude and the second magnitude, for a third period of time.
12. The circuit of claim 11, wherein the length of the second period of time is different than the length of the first period of time, and wherein the length of the third period of time is different than the length of the first period of time and the length of the second period of time.
13. The circuit of claim 11, wherein the threshold resistance is between the first resistance and the second resistance.
14. The circuit of claim 11, the circuitry configured to change the resistance of the memory element from the second resistance to the first resistance by setting the current through the memory element to have a fourth magnitude and a second sign, opposite the first sign, for a fourth period of time, a fifth magnitude, different than the fourth magnitude, and the second sign for a fifth period of time, and a sixth magnitude, different than the fourth magnitude and the fifth magnitude, for a sixth period of time.
15. The circuit of claim 11, wherein the second magnitude is greater than the first magnitude, and wherein the third magnitude is zero.

16. The circuit of claim 11, wherein the second magnitude is less than the first magnitude, and wherein the third magnitude is zero.

17. A circuit comprising: a memory array comprising a memory cell coupled to a bit line, a source line, and a word line, wherein the memory cell comprises a magnetic tunnel junction (MTJ) memory element coupled between the bit line and the source line; and circuitry coupled to the MTJ memory element through the bit line, the source line, and the word line, the circuitry configured to change a resistance of the MTJ memory element from a first resistance state to a second resistance state, different than the first resistance state, by providing a first current having a first magnitude and a first sign to the MTJ memory element for a first period of time, providing a second current having a second magnitude, different than the first magnitude, and the first sign to the MTJ memory element for a second period of time, which follows the first period of time, in response to the resistance of the MTJ memory element reaching a threshold resistance state that is between the first resistance state and the second resistance state, and providing a third current having a third magnitude, different than the first magnitude and the second magnitude, for a third period of time which follows the second period of time.

18. The circuit of claim 17, the circuitry configured to change the resistance of the memory element from the second resistance state to the first resistance state by providing a fourth current having a fourth magnitude and a second sign, opposite the first sign, to the MTJ memory element for a fourth period of time, providing a fifth current having a fifth magnitude, different than the fourth magnitude, and the second sign to the MTJ memory element for a fifth period of time which follows the fourth period of time, and providing a sixth current having a sixth magnitude, different than the fourth magnitude and the fifth magnitude, for a sixth period of time which follows the fifth period of time.

19. The circuit of claim 18, wherein the third magnitude is zero, the sixth magnitude is non-zero, and the sixth current has the second sign.

20. The circuit of claim 18, wherein the fourth magnitude is different than the first magnitude.